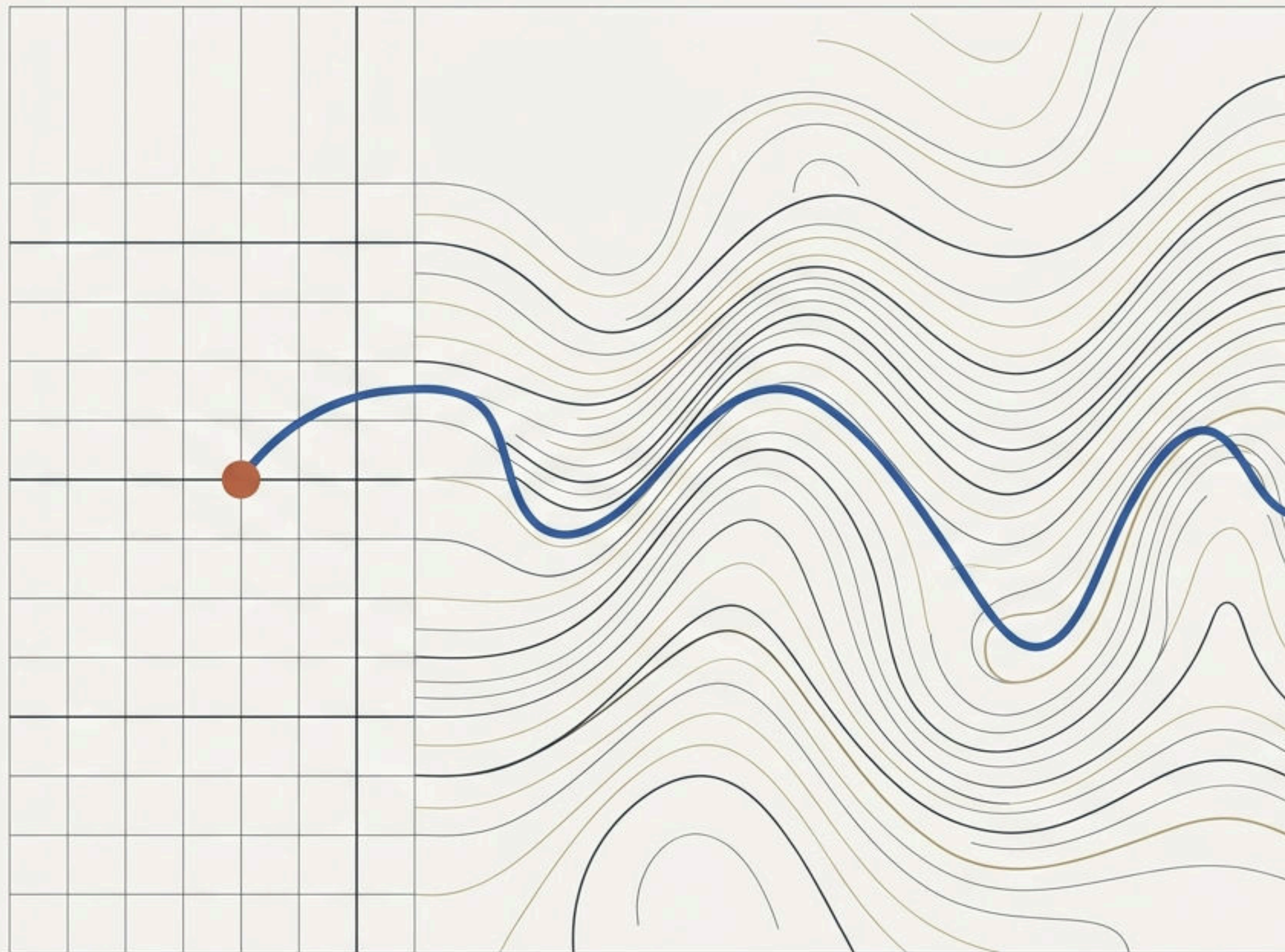


STATES WITHOUT HISTORIES

Latent Reasoning as Admissibility-Field Traversal

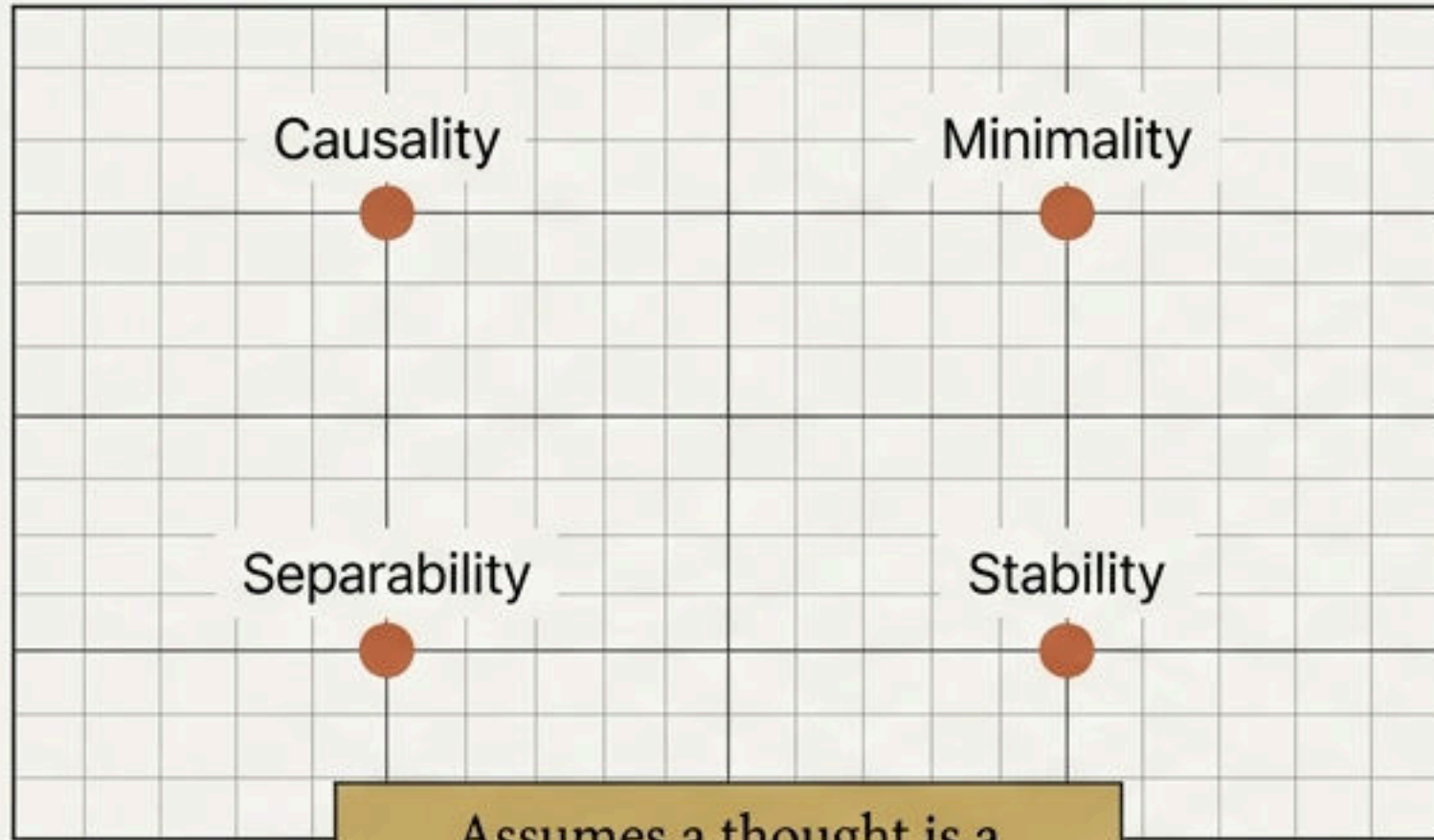
Moving beyond the “Thought as Location” paradigm to solve the separability collapse in Large Language Models.

Based on the June 2026 Flyxion
Independent Research report.



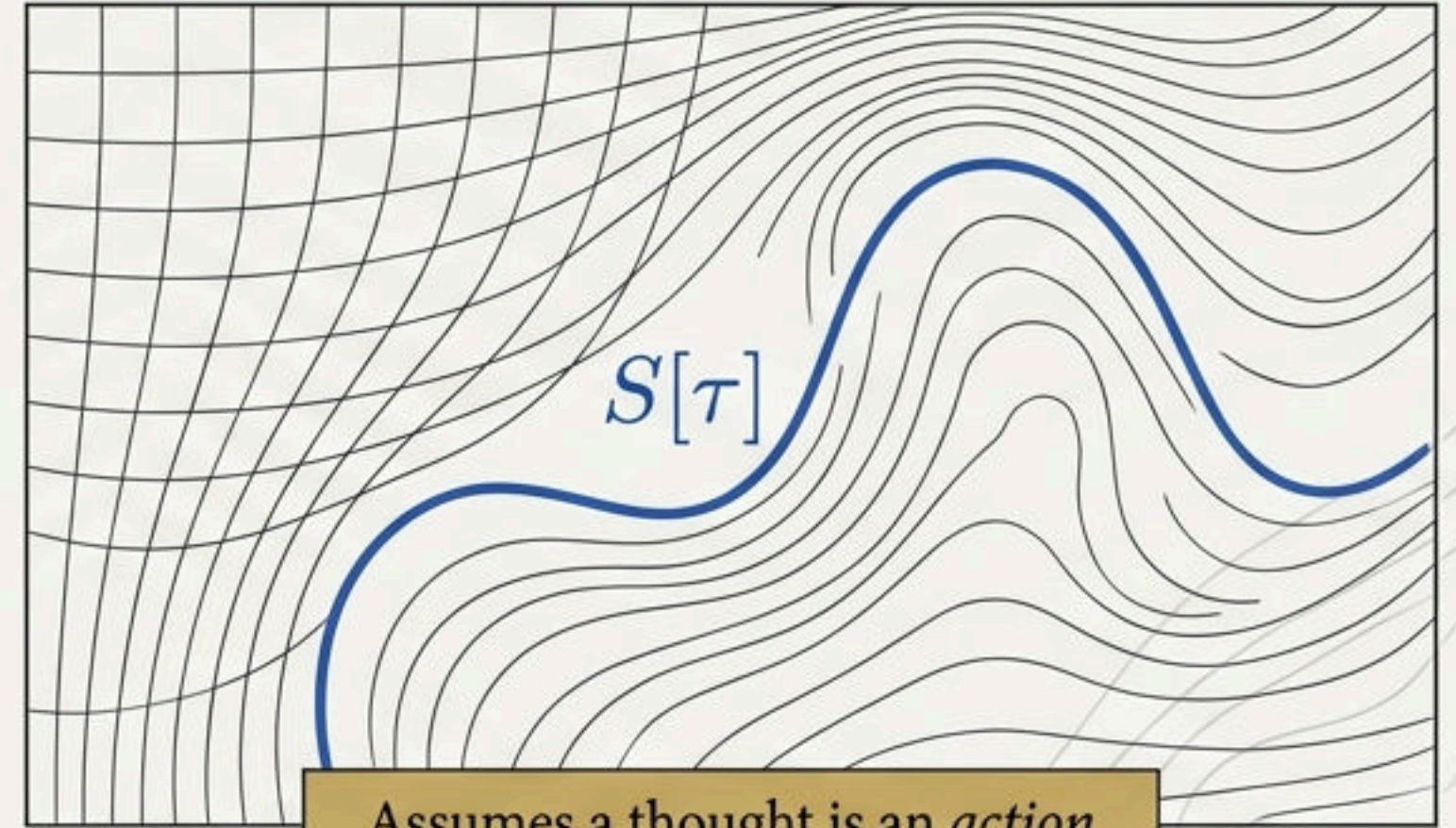
Current frameworks mistake the trajectory of thought for an instantaneous location

The State Ontology



Assumes a thought is a primitive location.

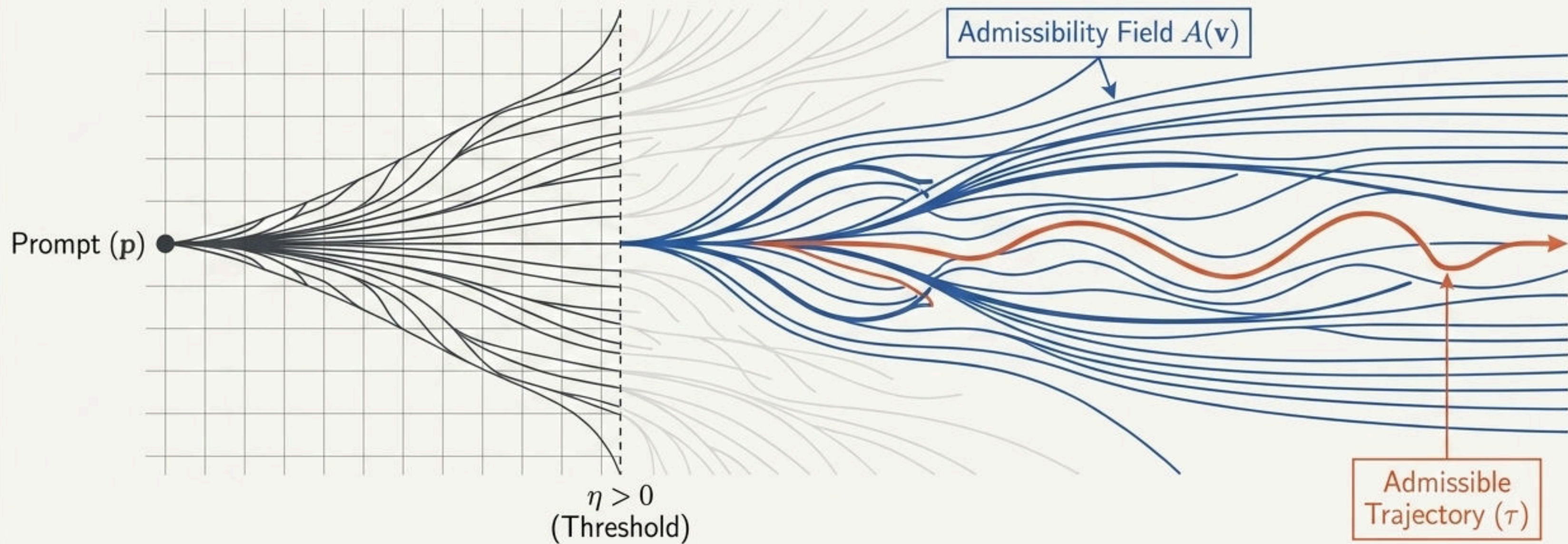
The Trajectory Ontology



Assumes a thought is an *action functional* across a path.

The Anomaly: Seddik and Fard (2026) proved that LLMs preserve task-level identity while collapsing instance-level identity.

The Diagnosis: This is not a failure of representation design. It is a fundamental mismatch between state-based measurement and path-dependent reasoning.



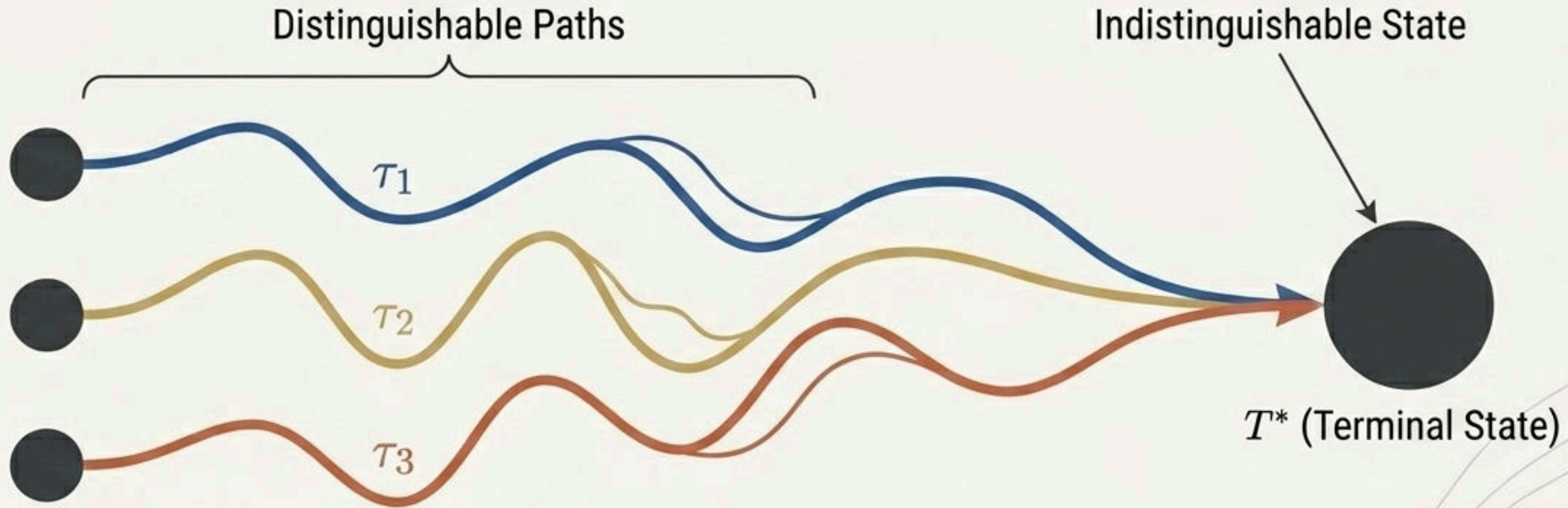
Definition 1: Admissibility Field $\mathcal{A}$

The set of next-token transitions receiving non-negligible probability mass.

Latent Induction

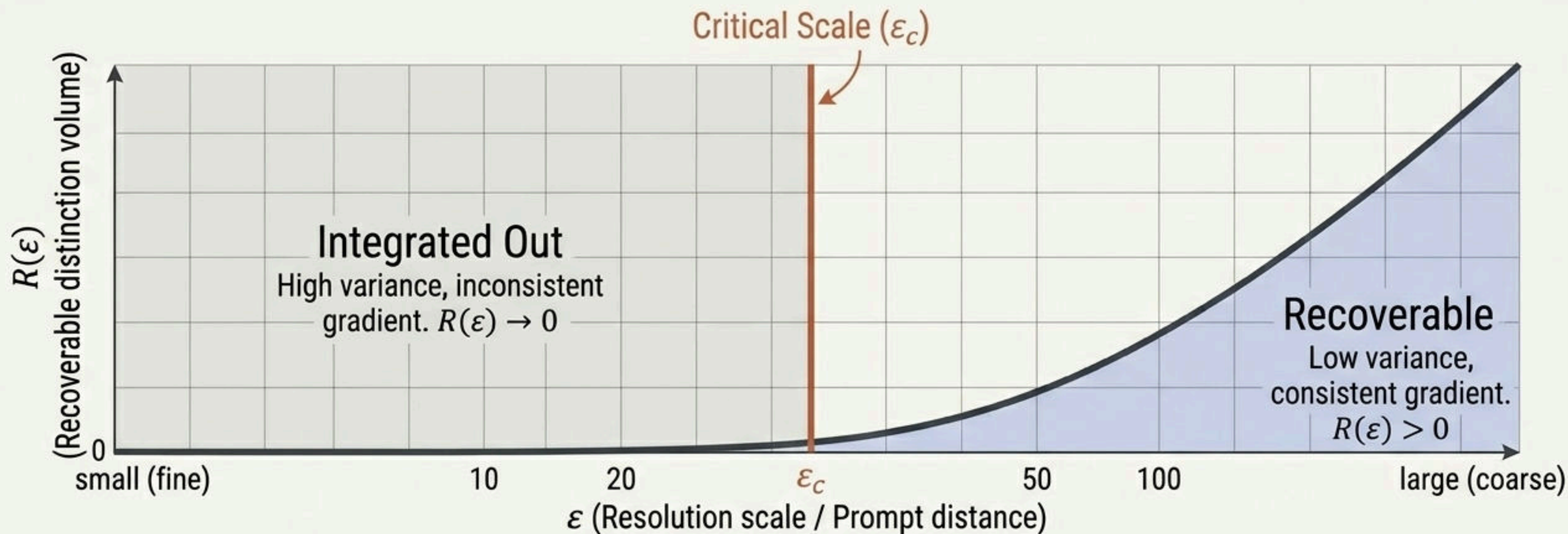
Token-level admissibility seamlessly lifts to latent-level equivalence without switching substrates. The prompt specifies the landscape; the traversal determines the region reached.

Knowing a latent state does not determine the reasoning trajectory



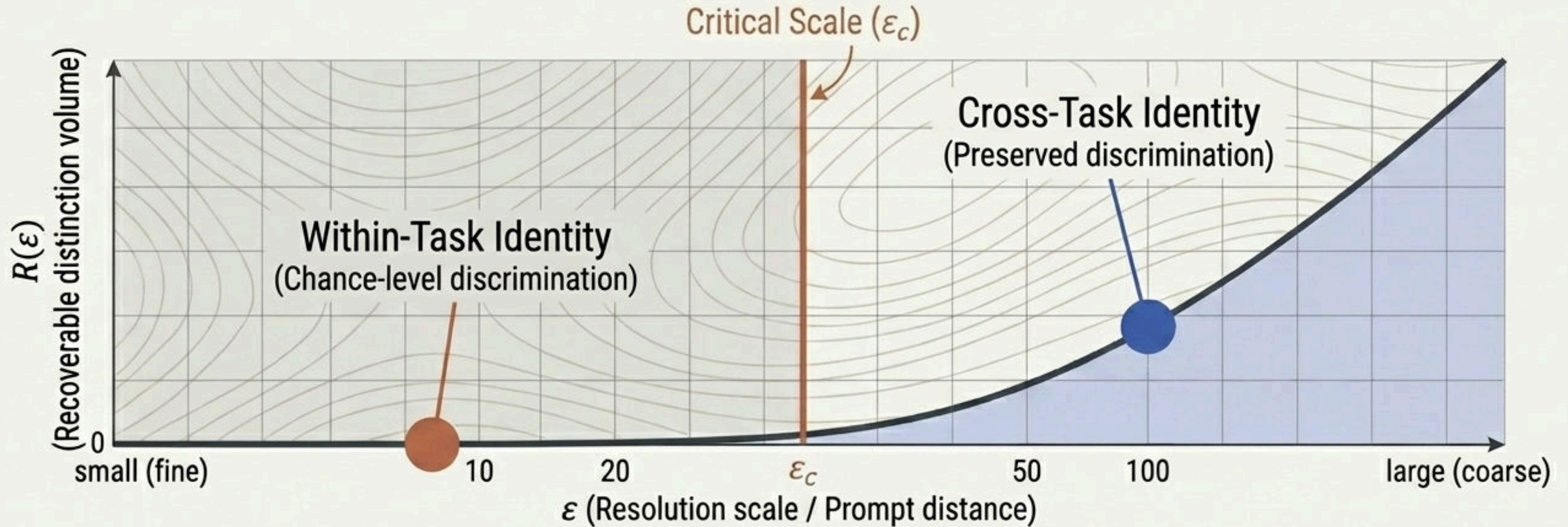
State-Trajectory Non-Equivalence (Proposition 2)	The Implication
Because the admissibility field is many-to-one, multiple distinct reasoning paths can reach the exact same terminal state.	Instance identity is distributed across the entire path. As step count increases, measuring a later state recovers less about the instance because the system has moved further from its starting conditions without aggregating that history.

Optimization acts as a renormalization operator that integrates out fine-grained distinctions



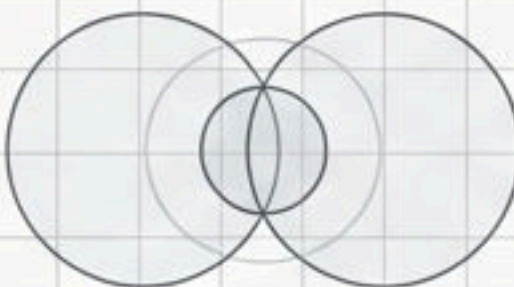
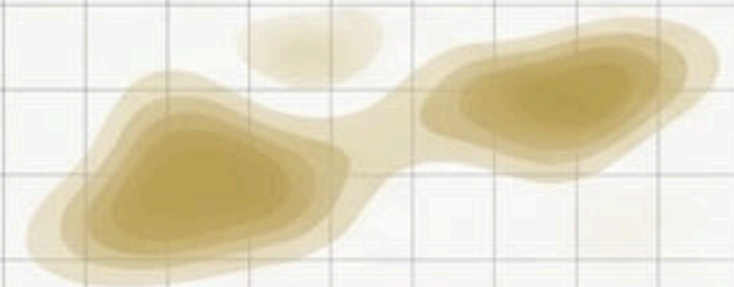
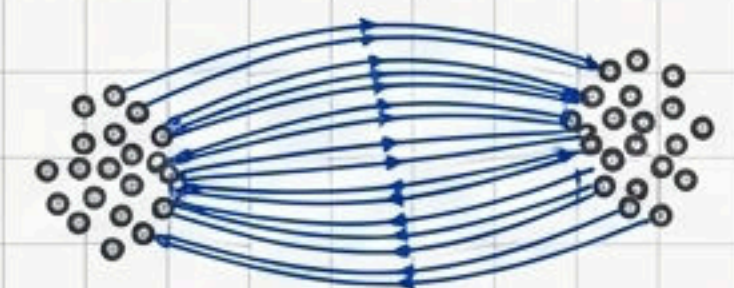
The Mechanism: Optimization pressure minimizes expected action. Distinctions with consistent gradient signals (low variance) survive. Distinctions with inconsistent signals (high variance) are deemed statistically costly and are erased by the representational capacity threshold.

The separability collapse is a structural theorem of optimization, not a design flaw



- Corollary 1 (Separability Collapse)** · We can mathematically recover Seddik & Fard's empirical hierarchy.
- Because $\epsilon_{\text{instance}} < \epsilon_c < \epsilon_{\text{task}}$, same-task discrimination converges to zero while cross-task discrimination remains positive.
 - The system expands into directions below ϵ_c , raising the participation ratio while dropping task purity.

Measuring admissibility requires continuation geometry, not point coordinates

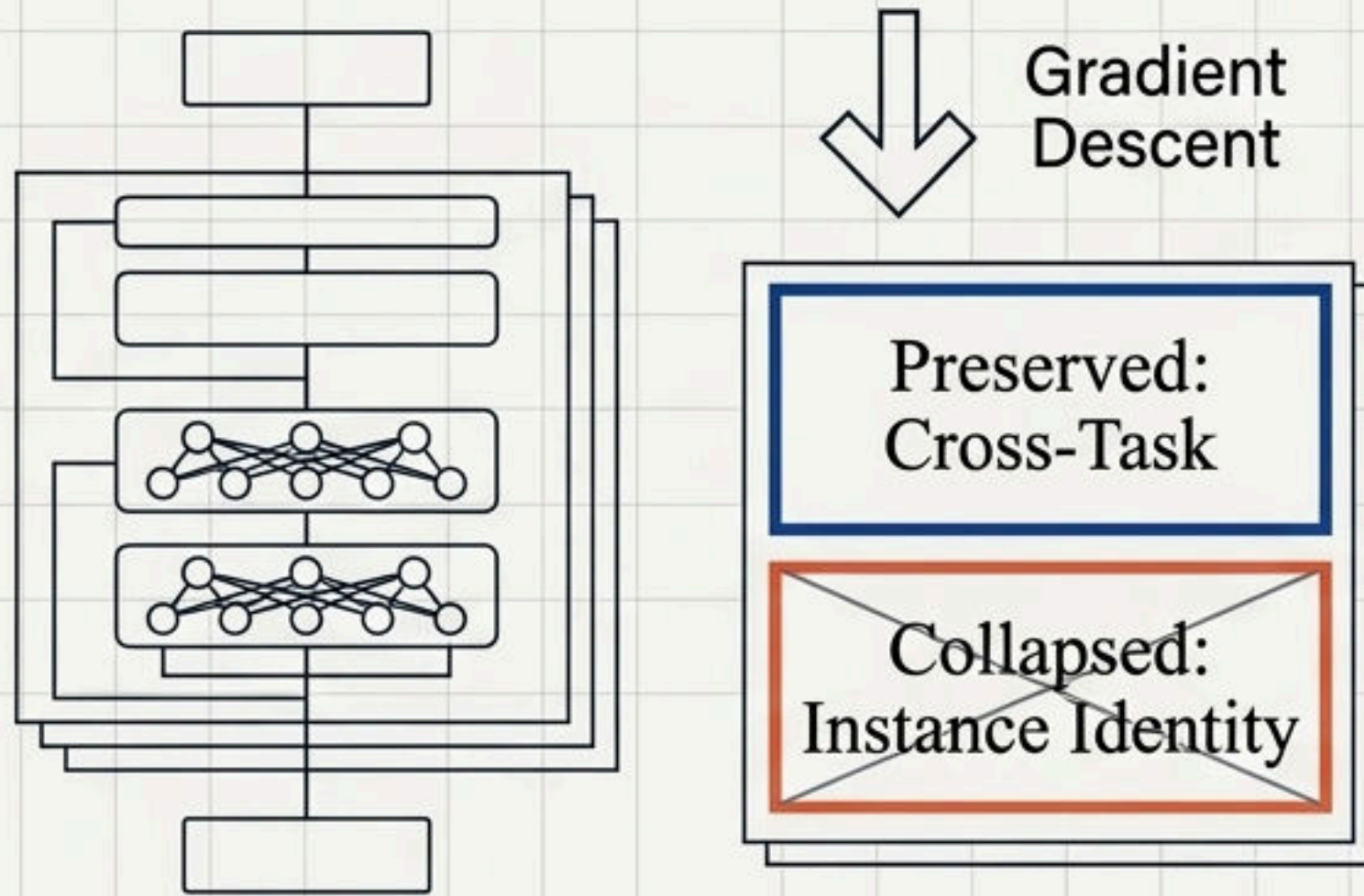
Hard Hausdorff	
	Baseline / Not Recommended. Beam search is mode-collapsing.
Soft-Hausdorff	
	Practical. Divergence metric based on generation probability.
Nucleus Wasserstein (D^W_A)	
	Theoretically Preferred. Measures optimal transport between the top-N token distributions.

The Shift

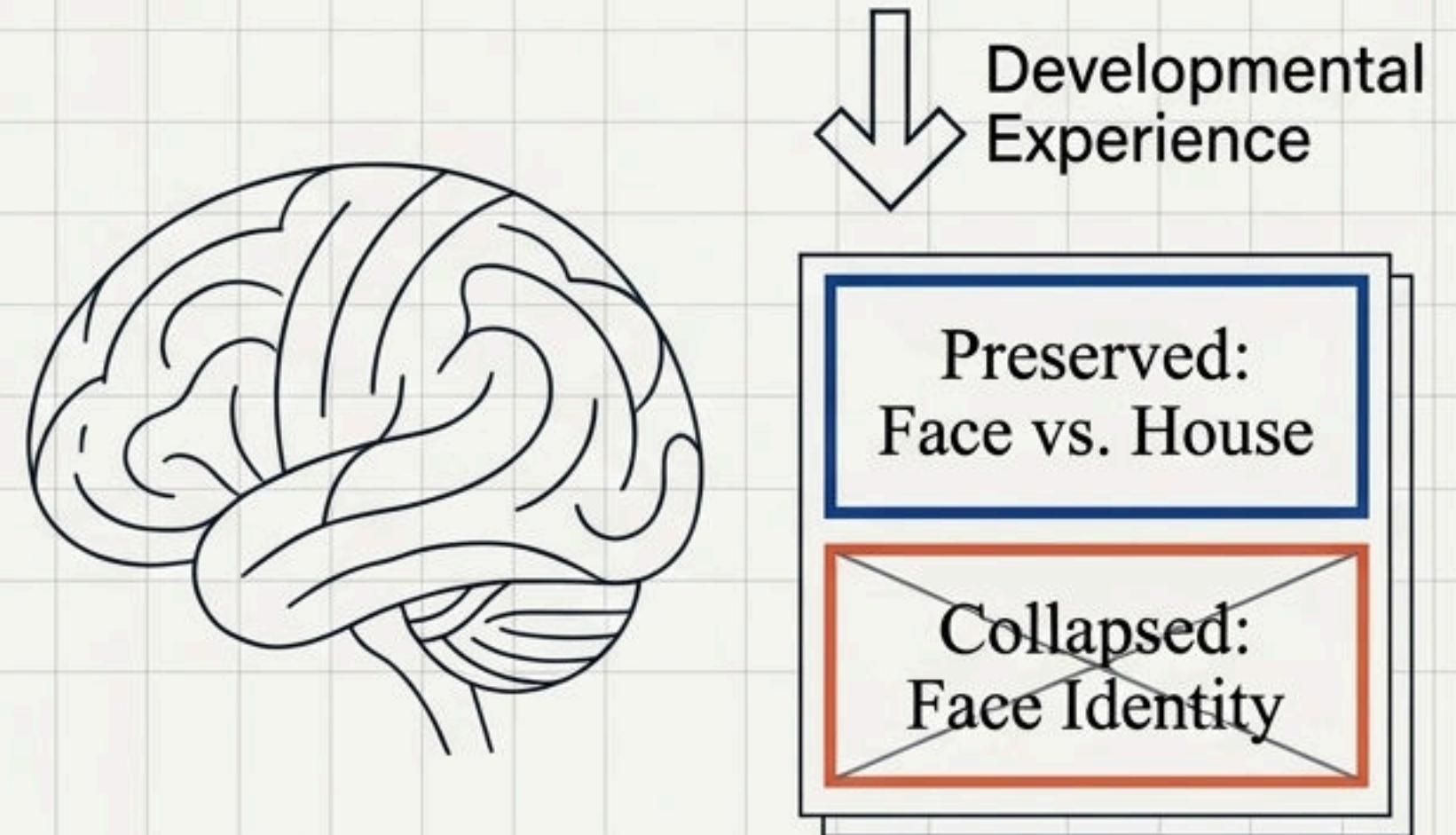
We must discard latent point coordinates and instead evaluate the shape of future distributions. By filtering out low-probability noise, the Nucleus Wasserstein metric perfectly scopes the genuine admissibility field.

The exact same two-level distinguishability collapse appears in human cortical networks

Silicon (LLMs)



Carbon (Human Cortex)



Cross-Domain Validation: Griffin et al. (2026) analyzed high-density EEG from 399 children. They found identical hierarchy patterns in cortical face processing in autistic children.

The Implication: ϵ_c is not a specific artifact of transformer architectures. It is a universal property of capacity-limited systems exposed to scale-uneven signals.

Biological experience drives the critical scale downward



Experience as Optimization

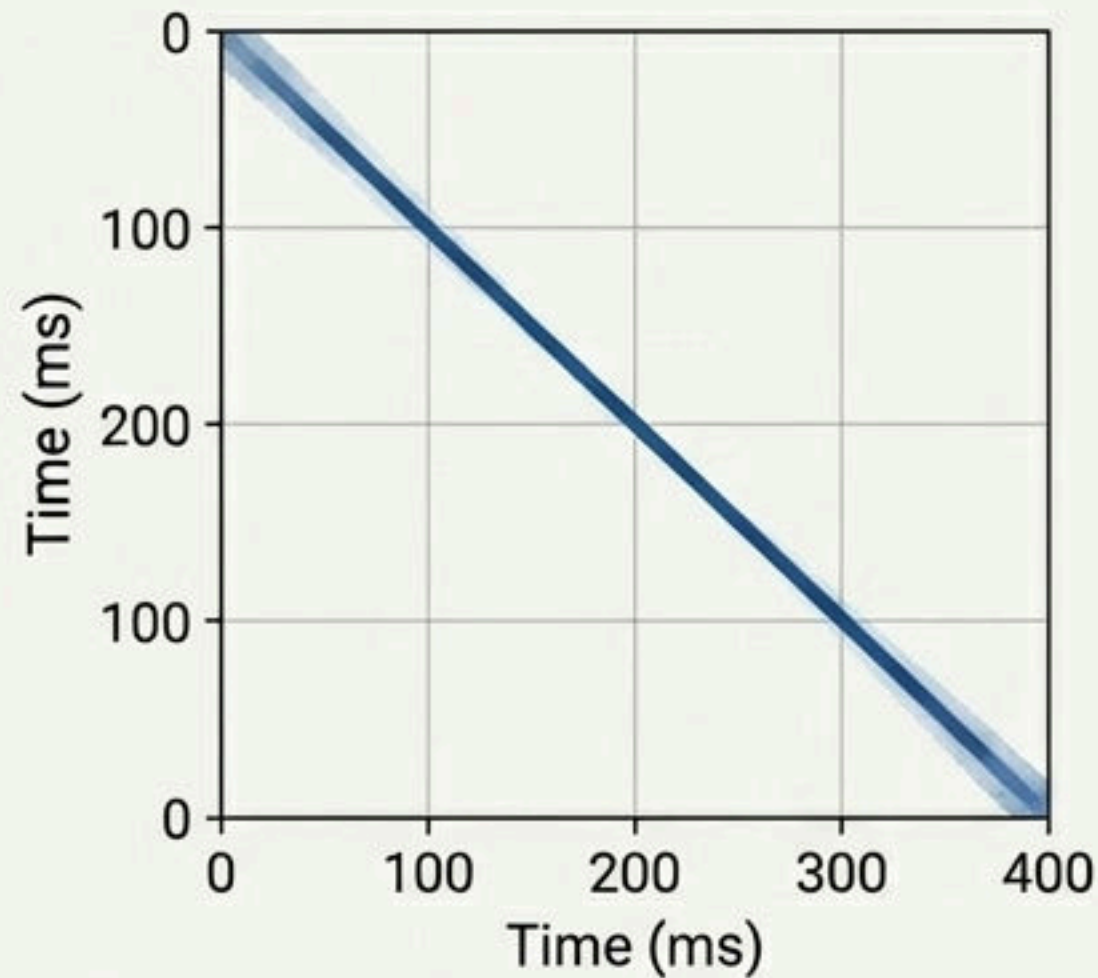
Consistent fine-grained input (face exposure) acts as a training signal, pushing ϵ_c downward to recover finer distinctions.

The Dissociation

Reduced social attention limits consistent signal. Without it, the critical scale remains elevated, leaving face identity permanently below the threshold of recoverable distinction.

Temporal generalization patterns prove fine-grained distinctions are trajectory-native

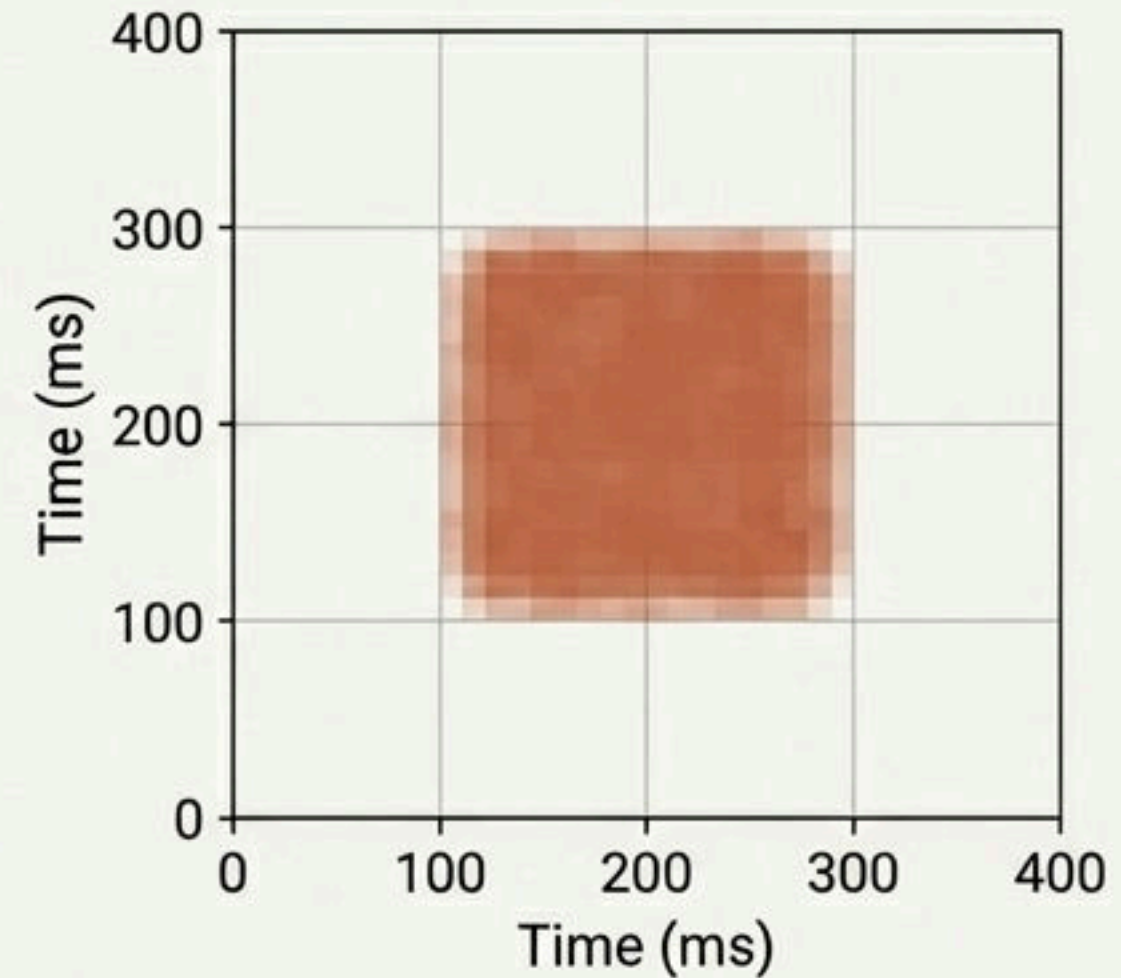
The Diagonal Signature



Neurotypical / Face-selective

Face-selective information relies on temporal sequences. It degrades over time from the initial stimulus, perfectly mirroring LLM step-degradation. A sequentially evolving trajectory.

The Square Signature



Autism / Identity-selective

The stable, fixed-point encoding indicates the system is not navigating a structured field. The trajectory has collapsed to a single point because there is no fine-grained admissibility structure to traverse.

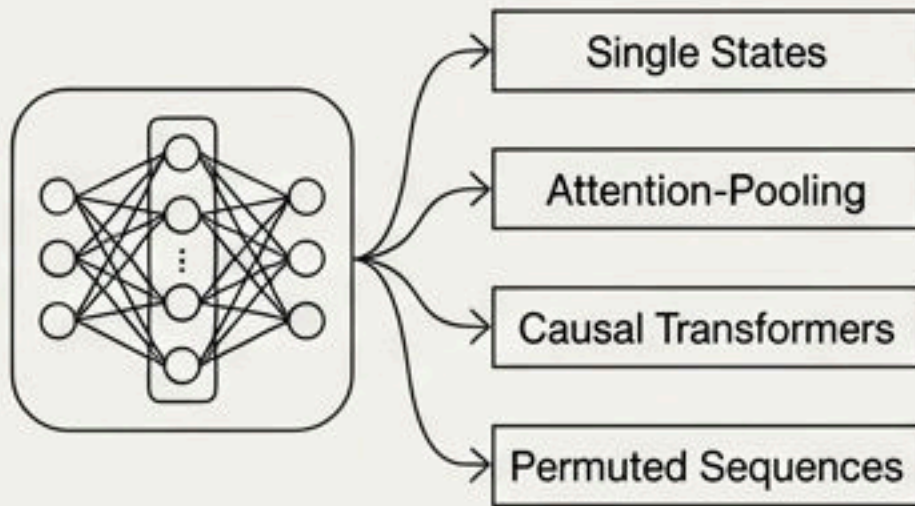
Diagnostic Matrix: Four competing hypotheses for the separability collapse

Hypothesis	Encoding Method	Linear Probe Recovery?	Unordered Aggregation Recovery?	Sequence Order Dependent?
H1: State	Single Latent	No	N/A	No
H2: Distributed State	Geometric Distribution	No	N/A	No
H3: Ensemble Averaging	Noise Reduction	No	Yes	No
H4: Trajectory Encoding	Temporal Structure	No	No	Yes

Seddik & Fard’s probe scaling eliminates H1 but leaves H2, H3, and H4 empirically indistinguishable at the single-state level.	Only rigorous sequence-permutation tests can isolate H4 as the true mechanism.
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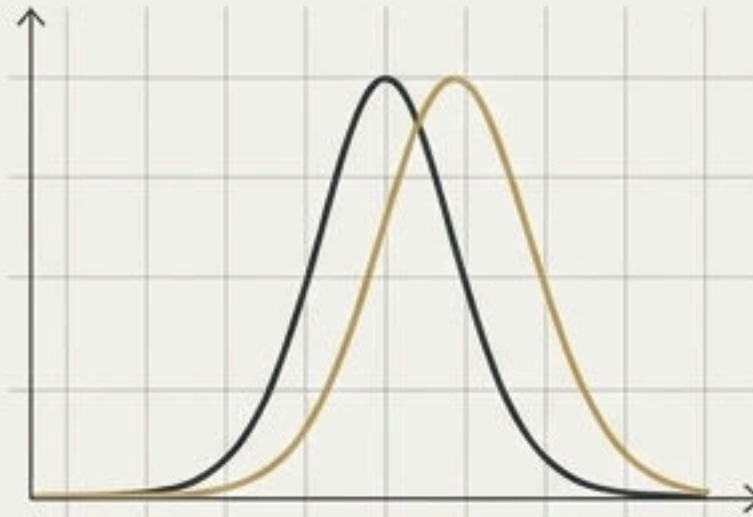
The Experimental Roadmap

Setup 1: Discriminator Matchup



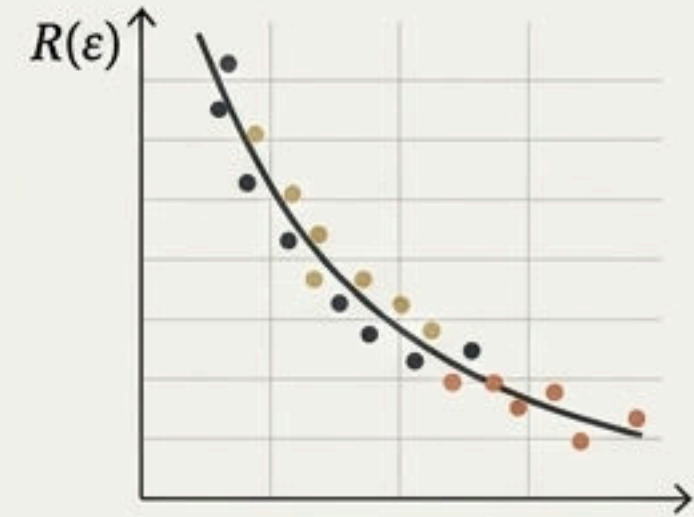
Compares capacity-matched single states vs. attention-pooling vs. causal transformers vs. permuted sequences to definitively isolate H4.

Setup 2: Geometric Divergence



Correlates soft-Hausdorff divergence against ground-truth answer divergence to prove continuation geometry outperforms point geometry.

Setup 3: Plotting the Curve



Bins same-task problem pairs by prompt distance to empirically locate the critical scale (ϵ_c) and calculate the slope of distinguishability recovery.

To validate the Trajectory Hypothesis, we must measure the path itself. If capacity-matched causal transformers applied to ordered sequences recover instance identity where single states fail, the state ontology must be retired.

If fine-grained identity is integrated out by training, no better state vector will ever find it

- The deepest claim is not that current representations fail. It is that **thought cannot be represented as a state variable** at the scale we are measuring.
- If the critical scale (ϵ_c) lies above the instance level, the training process has fundamentally integrated that distinction out of the state geometry.
- Correcting this requires abandoning the search for better latent point vectors. We must evaluate the right object: the **admissible trajectory** and the scale-dependent distinguishability it preserves.